

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester I Examination, 2016-17

Zoology

Paper: ZOB-101 — Animal Diversity & Animal Form and Function

Time: Three Hours

Full Marks: 35 + 35 = 70

Note: Answer three questions from Section-A and three questions from Section-B. Question No. 1 in both sections is compulsory. Use a separate answer book for each Section. The figures in the right-hand margin indicate marks.

SECTION – A
(Animal Diversity) (Marks: 35)

Note: Answer three questions including Question No. 1 which is compulsory.

Question 1. Answer the following parts:

- (a) State whether the following statements are True or False. Justify your answer in 2–3 sentences: [2 × 3 = 6]

- (i) Balanoglossus belong to class Pterobranchia of Phylum Hemichordata.
- (ii) Birds are glorified reptiles.
- (iii) Colloblasts are found in Cnidarians.

- (b) Distinguish between the following: [3 × 1 = 3]

- (i) Protostomes and Deuterostomes
- (ii) Urodela and Anura
- (iii) Pseudocoelomate and coelomate.

- (c) Define the following: [2 × 1 = 2]

- (i) Statocyst
- (ii) Symmetry.

Question 2. Answer the following parts: [7 + 5 = 12]

- (a) Classify phylum protozoa up to classes giving characteristic features and examples of each class.
- (b) Differentiate between cartilaginous and bony fishes.

Question 3. Answer the following parts: [7 + 5 = 12]

- (a) Classify phylum Mollusca up to classes giving characteristic features and examples of each class.

- (b) Describe water vascular system of Echinoderms.

Question 4. Write short notes on any three of the following:

[3 × 4 = 12]

- (a) Skull types in Reptiles
- (b) Cetaceans
- (c) Cyclostomes
- (d) Canal system of Poriferans.

SECTION – B
(Animal Form and Function) (Marks: 35)

Note: Answer three questions including Question No. 1 which is compulsory.

Question 1. Answer the following parts:

(a) State whether the following statements are True or False and give brief reasons in support of your answer: [2 × 4 = 8]

- (i) Longer the loop of nephron more water and ions can be reabsorbed.
- (ii) The presynaptic cell transmits signals from specialized enlarged regions of its axon's branches, called synaptic cleft.
- (iii) Claspers are accessory sex organ found in birds and mammals.
- (iv) Motor neurons carry commands from the brain or spinal cord to muscles and glands.

(b) Define the following terms: [1 × 3 = 3]

- (i) Suspension feeding
- (ii) Book lungs
- (iii) Sensory pathways.

Question 2. Answer the following parts: [6 + 6 = 12]

- (a) With the help of well labelled diagram explain the structure and functions of mammalian kidney. Describe the role of protonephridia in the process of excretion.
- (b) Describe the structure and function of gills in fishes. Explain the role of mammalian lung in gas exchange.

Question 3. Answer the following parts: [6 + 6 = 12]

- (a) Describe the patterns of nervous system in non-chordates with supporting well labelled diagram.
- (b) Give an account of the phonoreception in mammals.

Question 4. Write short notes on any three of the following: [3 × 4 = 12]

- (a) Intracellular digestion
- (b) Parthenogenesis
- (c) Accessory sex organs in mammals.

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